

AYMANE AIT DADS

Data Science Intern | AI-Driven Forecasting · Supply Chain Optimization · Predictive Modeling

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by fine-tuning a 30B-parameter model with LoRA SFT - demonstrating rigorous predictive modeling and analytical depth directly applicable to large-scale AI applications. Reduced manual analysis time by 60% at Orange Maroc by building Random Forest and K-Means pipelines on 100K+ network records, mapping customer performance profiles to 5 commercial service tiers via automated feature engineering. Brings strong Python programming, data preparation and analysis, Power BI visualization, and deep learning expertise to HP's data-driven supply chain optimization team. Aims to develop and implement AI-driven solutions for time series forecasting and inventory management within HP's global operations environment.

CORE COMPETENCIES

AI-Driven Solutions | Time Series Forecasting | Predictive Models | Supply Chain Optimization | Data Preparation and Analysis | Deep Learning Architectures | Large Language Models | Power BI Visualization

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built Random Forest and K-Means predictive models on **100K+ fiber-optic network records** to classify network quality across thousands of nodes, supporting AI-driven decision-making.
- Developed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, output adopted directly by the network optimization team for operational decisions.
- Reduced manual data preparation and analysis time by **60%** through automated feature engineering and model evaluation workflows built entirely in Python and Scikit-learn.
- Delivered stakeholder presentations translating model outputs into maintenance recommendations, communicating complex AI-driven findings to both technical and **non-technical audiences**.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation), building a diagnostic evaluation pipeline that revealed reasoning over-expansion - not incorrect reasoning - as the primary failure mode, and built custom cipher parsers and 8-bit operation solvers to produce verified chain-of-thought training data.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** on the private Kaggle leaderboard (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M parameters) with a custom MLP head, dual learning rates, and a 3-model ensemble with 10-view TTA trained on 250K images across 25+ generator types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark, applying FP16 mixed precision and deep learning best practices for out-of-distribution generalization in AI image classification.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

AI & Predictive Modeling: Python | PyTorch | Scikit-learn | Hugging Face Transformers | LoRA / PEFT | TensorFlow | vLLM

Data Analysis & Visualization: Pandas | NumPy | Power BI | Matplotlib | Seaborn | SQL | Feature Engineering

MLOps & Infrastructure: Git | Linux | FP16 / BF16 Mixed Precision | Docker (basic) | Kaggle GPU | REST APIs | PostgreSQL